

Formal Verification of Quantum Codes with Transversal Diagonal Gates in Lean 4

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arXiv:2510.20728 · github.com/LionSR/lean-qec

Transversal Gates & the SSLP Pipeline

Distance-2 Results

Lean 4 Formalization

Distance-3 Extension

Outlook

Summary

This Talk

Starting point: arXiv:2510.20728 catalogued 14,116 distance-2 codes with transversal diagonal gates via the SSLP pipeline.

Three new contributions:

1. **Lean 4 formalization** of the full SSLP pipeline — machine-checked proofs, zero sorry
2. **Analytical families:** one proof covers *infinitely many* codes
3. **Distance-3 generalization:** exact solutions for $((7, 2, 3))$ codes discovered by GPT-5.2, formally verified in Lean



Transversal Gates & the SSLP Pipeline

Transversal Diagonal Gates

Physical gate with angle vector $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{Z}_m^n$:

$$U(\mathbf{a}, m) = \bigotimes_{i=1}^n \begin{pmatrix} 1 & 0 \\ 0 & \omega_m^{a_i} \end{pmatrix}, \quad \omega_m = e^{2\pi i/m}$$

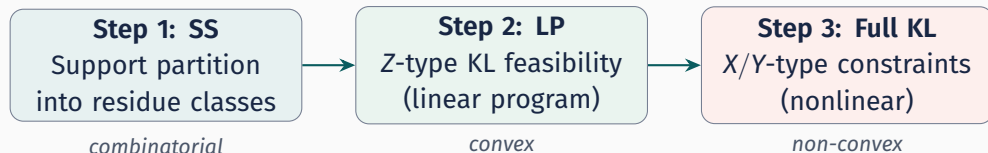
Action on computational basis: $U(\mathbf{a}, m) |x\rangle = \omega_m^{\langle \mathbf{a}, x \rangle} |x\rangle$ where $\langle \mathbf{a}, x \rangle = \sum_i a_i x_i$.

Define **residue classes** $C_b(\mathbf{a}) = \{x \in \{0, 1\}^n : \sum_i a_i x_i \equiv b \pmod{m}\}$.

Eigenstate condition

$$\text{supp}(|j_L\rangle) \subseteq C_{S_j}(\mathbf{a}) \implies U(\mathbf{a}, m) |j_L\rangle = \omega_m^{S_j} |j_L\rangle$$

Subset-Sum Linear Programming: systematic code construction.



Step 2 is an LP $\Rightarrow$ fast rejection of infeasible parameters.

For **distance 2**: our residue-shift screen makes Step 3 *automatic*.

For **distance 3**: we solve Step 3 via exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$.

Step 1: Subset-Sum Support Partition

Residue classes $C_b(\mathbf{a}) = \{x \in \{0,1\}^n : \sum_i a_i x_i \equiv b \pmod{m}\}$ partition $\{0,1\}^n$.

- **Disjoint:** $C_b \cap C_{b'} = \emptyset$ for $b \neq b' \Rightarrow$ automatic orthogonality
- **Logical action:** $U |j_L\rangle = \omega_m^{S_j} |j_L\rangle$
- **Projective order:**
 $\mathcal{O} = m / \gcd(m, S_1 - S_0, \dots)$

Example: $n=5, m=7$

$\mathbf{a}=(1, 1, 2, 2, 2), (S_0, S_1) = (0, 4)$

$C_0 = \{00000, 01111, 10111\}$

$C_4 = \{00011, 00101, \dots\}$ (6 elts)

$\mathcal{O} = 7 / \gcd(7, 4) = 7$

Step 2: Z-Type KL as Linear Program

Probabilities $p_{j,x} = |c_{j,x}|^2$ on support of $|j_L\rangle$.

Z-type KL: $\langle j_L | Z_i | j_L \rangle = t_i$ independent of j becomes

$$\sum_{x \in C_{S_j}} (-1)^{x_i} p_{j,x} = t_i \quad \forall i \in [n], \forall j$$

Linear in $p_{j,x}$ with constraints:

- $\sum_{x \in C_{S_j}} p_{j,x} = 1$
- $p_{j,x} \geq 0$

Standard LP feasibility.

Geometric view: sign vectors

$v(x) = ((-1)^{x_1}, \dots, (-1)^{x_n})$ are vertices.

Feasibility = nonempty $\bigcap_j \text{conv}(V_j)$.

Residue-Shift Screen: Distance 2 for Free

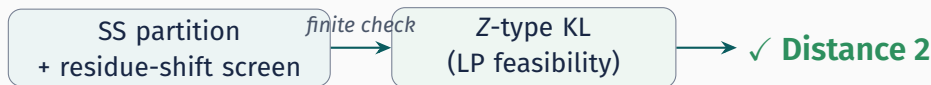
Flipping bit i shifts the residue by $\pm a_i$:

$$\langle \mathbf{a}, \mathbf{x} \oplus \mathbf{e}_i \rangle \equiv \langle \mathbf{a}, \mathbf{x} \rangle \pm a_i \pmod{m}$$

Residue-Shift Screen

$S_j - S_k \not\equiv \pm a_i \pmod{m}$ for all $i \in [n]$, all $j \neq k \implies$ union support has $d_H \geq 2$.

When the screen passes, **all X/Y-type KL vanish automatically.**



Binary Dihedral Symmetry ($K=2$)

Binary dihedral group BD_{2m} :

- Generated by $Z(2\pi/m)$ and $\hat{X} = -iX$
- $|BD_{2m}| = 4m$; $m=8 \Rightarrow$ transversal T
- Transversal $\bar{X} = X^{\otimes n}$: $|1_L\rangle = X^{\otimes n} |0_L\rangle$
- Support: $S_1 = \bar{S}_0$ (bitwise complements)
- Amplitudes: $c_{1,\bar{x}} = c_{0,x}$

BD angle condition

$$\sum_{j=1}^n a_j \equiv -1 \pmod{m}$$

Z-balance

Complement symmetry forces

$$\sum_{x \in S_0} (-1)^{x_i} p_x = 0 \text{ for all } i.$$

Only $|0_L\rangle$ needs specification.

Distance-2 Results

Systematic search via SSLP:

- $K \in \{2, 3, 4\}$, $n \leq 6$, $m \leq 18$
- Residue-shift screen for $d=2$
- LP solver + rational reconstruction

14,116

certified $((n, K, 2))$ codes

Highlights

- Orders $\mathcal{O} = 2$ through 18 at $n = 6$
- Mostly **nonadditive**
- Includes $K = 3, 4$ codes

Analytical families

Family I: $C_0 = \{0^n, 1^n\}$,
closed-form $p(0^n) = 1 - s/m$
✓ verified in Lean for all (n, m, s)

Analytical Families: One Proof, Infinitely Many Codes

Find a **closed-form pattern**, prove it *once*, instantiate everywhere.

Family I: the simplest codes

Support $C_0 = \{0^n, 1^n\}$ — just two basis states.

$$|0_L\rangle = \sqrt{p} |0 \cdots 0\rangle + \sqrt{1-p} |1 \cdots 1\rangle$$

Z-balance forces p **uniquely**:

$$p = 1 - s/m.$$

Valid for *all* $(n, m, \mathbf{a})$ passing the screen.

λ -families

Larger supports, free parameter λ :

- λ -Family II: $|C_0| = 4$
- λ -Family V: $|C_0| = 6$
- 5 families, each a **universally quantified** Lean theorem

One Lean proof $\Leftrightarrow$
infinitely many
verified codes

Worked Example: $((5, 2, 2))$ Code

$n=5, \mathbf{a}=(1, 1, 2, 2, 2), m=7, (S_0, S_1) = (0, 4)$

SS: $C_0 = \{00000, 01111, 10111\}$, C_4 : 6 elements

Screen: $4 \not\equiv \pm 1, \pm 2 \pmod{7}$ ✓

LP: $p_{00000} = 3/7, p_{01111} = p_{10111} = 2/7$
 $\langle Z_i \rangle = 3/7$ ($i=1, 2$), $-1/7$ ($i=3, 4, 5$)

Lean: one-line proof

```
def myCode : ExampleData where
  n := 5; m := 7; K := 2
  a := ![1, 1, 2, 2, 2]
  S := ![0, 4]
  supp := ![supp0, supp1]
  prob := ![prob0, prob1]

theorem verified : myCode.OK :=
  by native_decide
```

SS + screen + LP $\Rightarrow$ certified $((5, 2, 2))$ code, order $\mathcal{O} = 7$. Build: < 1 s.

Lean 4 Formalization

What is Lean 4?

A **proof assistant** based on dependent type theory:

- Every proof is a program; the **kernel** type-checks it
- Compiles $\Rightarrow$ proof is correct. No trust required.
- **Mathlib**: 180K+ formalized theorems

Key tactic: `native_decide`

- Proposition is `Decidable`? Compile to native code $\rightarrow$ run $\rightarrow$ true
- Kernel accepts: proof complete
- “All 210 KL constraints hold?” $\rightarrow$ true $\rightarrow$ QED

Landmark projects

- Freiman–Ruzsa conj. (Tao)
- Liquid Tensor Experiment
- Sphere eversion
- **This work**: quantum codes

Why Lean for QEC?

14K codes: too many for humans.

AI-found solutions: trust but verify.

Formal proof = **certainty**.

Lean 4 Codebase: lean-qec

Core Library (ss/)

BitString.lean
Basic.lean
DiagonalAction.lean
ResidueShiftScreen.lean
ZTypeKL.lean
BDSymmetry.lean
Verify.lean
FamilyI/ (6 files)
LambdaV2/ (5 families)

Distance-3 Modules

Pauli.lean
FullKL.lean
QSqrt23i.lean
QSqrt235i.lean
FullKLEval.lean
Distance3Verify.lean
BD16Defs.lean

Codebase

~4,300 lines, ~30
modules
0 sorry, 0 warnings

Examples

Examples/D2/
Ex522.lean ((5,2,2))
Ex622.lean ((6,2,2))
Examples/D3/
BD16v1.lean through
BD16v10.lean (12 instances)
NoGo/ (2 no-go proofs)

Build

3,092 jobs
0 errors · 0 warnings

Formalization: Key Theorems

Dact_eq_global_of_SS

$$\text{supp}(|j_L\rangle) \subseteq C_{S_j} \Rightarrow U |j_L\rangle = \omega_m^{S_j} |j_L\rangle$$

no_hamming1_neighbor_of_screen

Screen condition $\Rightarrow d_H(x, y) \geq 2$
across classes
 $\Rightarrow$ all X/Y-type KL vanish

ExampleData.OK

Bundles four checks:

1. SS support condition
2. Residue-shift screen
3. Probability normalization
4. Z-type KL matching

All Decidable $\Rightarrow$
native_decide in $< 1s$

Technique: BitString $n = \text{Fin } n \rightarrow \text{Bool}$ has 2^n elements. All quantifiers over finite support are decidable $\Rightarrow$ Lean compiles to native code, executes, returns true $\Rightarrow$ kernel accepts as proof.

Distance-3 Extension

The Distance-3 Barrier

Why was distance 2 “easy”?

- Residue-shift screen kills all X/Y-type KL automatically
- Only Z-type survives $\Rightarrow$ linear in probabilities $p_x \geq 0$
- **No complex amplitudes needed**

Why is distance 3 harder?

- Must check all weight- ≤ 2 Paulis: $Z_i Z_j, X_i X_j, X_i Y_j, \dots$
- X, Y **flip bits** $\Rightarrow$ couples different basis states
- KL becomes **quadratic** in $c_x \in \mathbb{C}$
- **Relative phases matter**

The gap: distance-2 = convex optimization over $\mathbb{R}_{\geq 0}$.

Distance-3 = nonlinear system over $\mathbb{C}$. At $n=7$: 210 weight- ≤ 2 Pauli constraints.

Core challenge: solve 210 nonlinear constraints over $\mathbb{C}$, then verify with mathematical certainty.

Distance 3: Three Key Ideas

1. Most constraints vanish.

$$S_0 \cap (S_0 \oplus z_P) = \emptyset$$

$\Rightarrow$ sum is empty $\Rightarrow 0$.

BD16v1: 25 / 210

> 88% are automatic.

2. Exact number field.

Amplitudes in
 $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, i)$

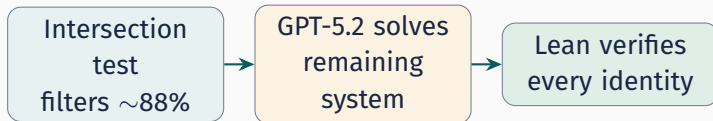
= 16 rational
coefficients.

DecidableEq, no floats.

3. AI + Lean.

GPT-5.2 solves the
surviving constraints
symbolically.

Lean verifies *exactly*.



Why Most KL Constraints Vanish

$\langle 0_L | P | 0_L \rangle$ sums over pairs where x and $x \oplus z_P$ are *both* in S_0 .



Disjoint $\Rightarrow$ empty sum $\Rightarrow$ automatically 0

$S_0 \cap (S_0 \oplus z_P) = \emptyset \implies \langle 0_L | P | 0_L \rangle = 0.$

Why so effective

S_0 is **sparse**: $|S_0| = 7$ out of $2^7 = 128$. Random shifts almost never land back inside S_0 . Only 25 / 210 survive (BD16v1).

In Lean: `klIsAutomatic` is Decidable — checked by `native_decide`.

From Constraints to Solutions: GPT-5.2

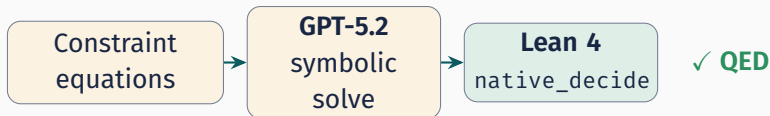
After intersection test, the surviving constraints form a **quadratic system** in $c_x \in \mathbb{C}$.

The problem:

- Unknowns: c_x for each $x \in S_0$
- $\sum_x \bar{c}_x c_{x'} \cdot (\text{phase}) = 0$
- Non-convex $\Rightarrow$ hard for standard solvers

GPT-5.2's approach:

- Given S_0 , probabilities, constraint eqns
- Exploits symmetries, groups terms
- Returns **closed-form** amplitudes in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, i)$



Exact Arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, i)$

$m=8 \Rightarrow \omega_8 = \frac{1+i}{\sqrt{2}}$. Amplitudes involve $\sqrt{2}, \sqrt{3}$ from normalization, and $\sqrt{5}$ appears in remaining BD_{16} solutions (via $\sqrt{10}$).

Representation (qsqrt235i)

Each element is $u + v\sqrt{5}$ with
 $u, v \in \mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$
(16 rational coefficients total).

Multiplication closed.

Equality *decidable* (rational coefficients).

No floating point. Every operation exact over $\mathbb{Q}$. `DecidableEq` $\Rightarrow$ `native_decide` verifies each KL identity.

BD16v1 amplitudes (GPT-5.2)

$$c_0 = \frac{1}{4} \quad c_5 = \frac{1}{4}$$

$$c_1 = c_2 = \frac{\sqrt{3}}{4} \quad c_6 = \frac{i}{2}$$

$$c_3 = c_4 = \frac{\sqrt{2}}{4}$$

In Lean: $c_1 = (0, 0, \frac{1}{4}, 0, 0, 0, 0, 0)$ ✓

Two-Layer Verification Architecture

Layer 1 (combinatorial, on probabilities)

SS + BD angle + normalization + Z-

balance: $\sum (-1)^{x_i} p_x = 0 \quad \forall i$

↓ *pass*

Layer 2 (amplitude-level, on $c_x \in \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, i)$)

All weight- ≤ 2 KL matrix elements = 0

(non-automatic only; rest filtered by intersection test)

Both layers are **fully decidable** — verified by `native_decide`.

No sorry, no axioms. Build time per code: 2–11 s.

BD16 Example: $((7, 2, 3))$ with $\mathbf{a} = (1, 2, 2, 2, 2, 3, 3)$

$m=8$ (transversal T), $|S_0| = 7$.

$$4|0_L\rangle = |0000000\rangle + \sqrt{3}|0111100\rangle + \sqrt{3}|1001110\rangle + \sqrt{2}|1010101\rangle + \sqrt{2}|1011001\rangle + |1110010\rangle + 2i|0101010\rangle$$

```
def bd16Data : Distance3Data 7 8 where
  a := ![1, 2, 2, 2, 2, 3, 3]₀
  b := 0₀
  supp := {s0, s1, s2, s3, s4, s5, s6}₀
  prob := ... -- (1,3,3,2,2,1,4)/16

theorem layer1 : bd16Data.Layer10K
  := by native_decide
theorem layer2 : AllKLSatisfied ₀supp ₀amp 2
  := by native_decide
```

Check	Result
SS + BD + norm	✓ native_decide
Z-balance	✓ native_decide
Non-auto KL	25 / 210
All 25 = 0	✓ native_decide

Complete $d \geq 3$ proof

Build: 2.6 s

12 Verified BD16 ((7, 2, 3)) Instances in Lean

All instances: $n=7$, $K=2$, $m=8$ (transversal T), verified Layer10K + AllKLSatisfied (weight- ≤ 2).

File	a	Lean
BD16v1	(1, 2, 2, 2, 2, 3, 3)	✓
BD16v2	(1, 1, 2, 2, 2, 2, 5)	✓
BD16v3a	(0, 1, 2, 2, 3, 3, 4)	✓
BD16v3b	(0, 1, 2, 2, 3, 3, 4)	✓
BD16v3c	(0, 1, 2, 2, 3, 3, 4)	✓
BD16v4	(1, 1, 1, 1, 3, 3, 5)	✓

File	a	Lean
BD16v5	(1, 1, 1, 1, 3, 4, 4)	✓
BD16v6a	(1, 1, 1, 2, 2, 4, 4)	✓
BD16v7	(1, 1, 2, 2, 2, 3, 4)	✓
BD16v8	(1, 1, 2, 2, 3, 3, 3)	✓
BD16v9	(1, 1, 1, 2, 3, 3, 4)	✓
BD16v10	(1, 1, 1, 2, 2, 3, 5)	✓

These 12 Lean files cover all 10 feasible BD16 angle vectors (v3 has three isolated solutions). Continuous families are verified at fixed representative parameters.

What the Surviving Constraints Look Like

After the intersection test, ~ 25 – 150 non-automatic Pauli strings remain.

Diagonal ($Z_i Z_j$)

$$\langle 0_L | Z_i Z_j | 0_L \rangle = \sum_{x \in S_0} (-1)^{x_i + x_j} |c_x|^2$$

Linear in probabilities $p_x = |c_x|^2$.

Already verified in **Layer 1**.

Key simplification

BD symmetry $|1_L\rangle = X^{\otimes n} |0_L\rangle$

$\Rightarrow$ off-diagonal eqns **pair up**:

$$\langle 0 | P | 0 \rangle = 0 \Leftrightarrow \langle 1 | P | 1 \rangle = 0$$

Only $|0_L\rangle$ amplitudes needed.

Off-diagonal ($X_i X_j, X_i Y_j, \dots$)

$$\langle 0_L | P | 0_L \rangle = \sum_{\substack{x \in S_0 \\ x \oplus z_P \in S_0}} \bar{c}_x c_{x \oplus z_P} \omega^{f(x, P)}$$

Quadratic in amplitudes $c_x \in \mathbb{C}$.

Phases $\omega^{f(x, P)}$ depend on x_P bits.

BD16v1: 7 nonzero amplitudes, 25 non-automatic constraints. Probabilities fixed by Layer 1 $\rightarrow$ only phases remain.

The BD_{16} Solution Landscape

All 12 SS-LP-passing BD_{16} $((7, 2, 3))$ angle vectors:

Angle vector $\mathbf{a}$	$ S_0 $	Non-auto	Solution type	Status
(1, 2, 2, 2, 2, 3, 3)	18	83	2-param continuous	Lean ✓
(1, 1, 2, 2, 2, 2, 5)	14	97	192-elt discrete	Lean ✓
(0, 1, 2, 2, 3, 3, 4)	16	68	3 isolated solutions	Lean ✓
(1, 1, 1, 1, 3, 3, 5)	13	81	continuous	Lean ✓
(1, 1, 1, 1, 3, 4, 4)	12	85	128-elt discrete	Lean ✓
(1, 1, 1, 2, 2, 4, 4)	16	83	uniform magnitude	Lean ✓
(1, 1, 2, 2, 2, 3, 4)	16	83	phase-only + families	Lean ✓
(1, 1, 2, 2, 3, 3, 3)	16	85	continuous ($\sqrt{10}$ coeffs)	Lean ✓
(1, 1, 1, 2, 3, 3, 4)	16	85	isolated ($\sqrt{10}$ coeffs)	Lean ✓
(1, 1, 1, 2, 2, 3, 5)	14	99	isolated ($\sqrt{5}$ coeffs)	Lean ✓
(1, 1, 1, 2, 2, 2, 6)	10	115	proven infeasible	
(0, 1, 1, 2, 3, 3, 5)	14	78	proven infeasible	

$|S_0|$, non-auto from Lean #eval. 10 feasible, all 10 in Lean (exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, i)$).

Lean No-Go Proofs for the Remaining 2 BD_{16} Vectors

Two angle vectors pass SS-LP but admit **no** complementary full-KL solution. We formalize **infeasibility** by proving a small KL subsystem already contradicts.

- Work under $|1_L\rangle = X^{\otimes 7} |0_L\rangle$ and amplitudes $c_s \in \mathbb{C}$.
- Prove **non-existence** statements (not just numerical optimization).
- Split the job:
 1. KL subsystem $\Rightarrow$ explicit polynomial equations
 2. equations $\Rightarrow$ contradiction by algebra/inequalities

```
-- Sec 16
theorem no_solution :
  Not (Exists fun c : Fin 10 ->
    Complex =>System c) := ...

-- Sec 17.6
theorem no_KLSubsys :
  Not (Exists fun c : Fin 14 ->
    Complex =>KLSubsys c) := ...
```

Lean files

Examples/D3/NoGo/BD16_1112226.lean

No-go #1: $\mathbf{a} = (1, 1, 1, 2, 2, 2, 6)$ (Sec. 16)

A tiny KL subsystem suffices

Unknowns: $c_0, \dots, c_9 \in \mathbb{C}$ on a $|S_0| = 10$ residue support.

Subsystem: **17 equations** = norm + 7 diagonal (Z_i) + 9 off-diagonal.

Step 1: solve the diagonal system

Let $p_k := |c_k|^2$ and set $u := p_7, v := p_9$.
The diagonal equations force

$$p_0 = p_1 = p_2 = p_3 = \frac{1}{16}, \quad p_6 = \frac{1}{4} - u, \quad p_8 = \frac{1}{4} - v, \quad p_4$$

Step 2: phase-gauge contradiction

Off-diagonal equations imply three product relations, e.g.

$$\bar{b}a = -\bar{d}c, \quad \Im(\bar{b}a) = 0.$$

These force $a, b, c, d, e, f \neq 0$.

Global phase invariance lets us fix a gauge with $b \in \mathbb{R}_{>0}$. Then all six special amplitudes become real, and the subsystem reduces to an inconsistent real algebraic system.

No-go #2: $\mathbf{a} = (0, 1, 1, 2, 3, 3, 5)$ — 19 quadratic equations

Here $|S_0| = 14$. We encode a sufficient KL subsystem as **19 explicit equations** (E1)–(E19) in $c_0, \dots, c_{13} \in \mathbb{C}$.

Diagonal part

- (E1) normalization: $\sum_k |c_k|^2 = 1$
- (E2)–(E8) seven Z_i constraints
- (E9) one Y_1 diagonal constraint (imaginary parts)

Off-diagonal part

- (E10)–(E12): X_2, X_2Z_1, X_2Z_3
- (E13)–(E15): X_3, X_3Z_1, X_3Z_2
- (E16)–(E17): $X_1X_2, X_1X_2Z_1$
- (E18)–(E19): $X_1X_3, X_1X_3Z_1$

Lean: `Examples/D3/NoGo/BD16_0112335_Equations.lean` proves `System.no_solution`.

Formalizing the missing step: $KL \Rightarrow (E1)-(E19)$

We do *not* assume (E1)–(E19) magically: we derive them from a KL subsystem on the explicit SS support S_0 .

Mechanization idea

- Hard part in Lean is bookkeeping: which pairs $(x, x \oplus z_P)$ contribute?
- We precompute intersections like $S_0 \cap (S_0 \oplus z_P)$ by `native_decide` and register them as `simp` lemmas.
- Then each KL matrix element reduces to a short explicit sum in the c_k .

```
structure KLSubsys (c : Fin 14 ->
  Complex) : Prop where
  norm : ...
  z1 : diagMatElt S0 (coeff c) b0 e1 = 0
  ...
  x13_ze1 : offdiagMatElt S0 (coeff c)
    x13 e1 = 0

theorem KLSubsys.toSystem (h : KLSubsys
  c) :
  System c := by
  -- simp + linarith
```

Core contradiction in the 19-eq no-go (Sec. 17.6)

From the Z -system we get **pair sums**:

$$p_0 + p_7 = \frac{1}{16}, \quad p_1 + p_8 = \frac{1}{16}, \quad p_2 + p_9 = \frac{1}{16}, \quad p_3 + p_{10} = \frac{1}{8}, \quad p_4 + p_{11} = \frac{3}{16}, \quad p_5 + p_{12} = \frac{3}{16}, \quad p_6 + p_{13} = \frac{5}{16}.$$

Off-diagonal equations force $p_{11} = p_{12} = \frac{3}{5} p_{13}$ and bounds $\tau := p_{13} \in [\frac{15}{176}, \frac{5}{22}]$.

Split the Y_1 equation

Write (E9) as $U + T = 0$:

$$U := \sum_{k=0}^3 \Im(\bar{c}_k c_{k+7}), \quad T := \sum_{k=4}^6 \Im(\bar{c}_k c_{k+7}).$$

Upper vs lower bounds

Using

$$|\Im(\bar{x}y)| \leq \frac{\|x\|^2 + \|y\|^2}{2},$$

pair sums give $|U| \leq \frac{5}{32}$.

But $\tau \geq \frac{15}{176}$ forces $|T| \geq \frac{3}{16} = \frac{6}{32}$.

Contradiction: $|T| = |U|$ is impossible.^{29/33}

Outlook

Outlook: Batch Verification of All 14,116 Codes

The distance-2 catalogue has 14,116 $((n, K, 2))$ codes. Currently: individual Lean files for representative examples.

Scaling strategy

1. Python exports each code's data as a `.lean` file (support, probabilities, angle vector)
2. Lean's `ExampleData.OK` bundle checks SS + screen + normalization + Z-KL
3. `native_decide` proves each in < 1 s
4. `lake build` compiles all 14K files

Already demonstrated

- `Ex522.lean`, `Ex622.lean`:
fully verified via `native_decide`
- Build time: 0.3–0.8 s each
- At 1 s/code: ~ 4 hours total
(embarrassingly parallel)

14,116 machine-checked proofs: within reach

Outlook: Analytical Family II & Beyond

Family II: 4-element support

$$C_0 = \{0^n, 1^n, e_1 \bar{e}_1, \bar{e}_1 e_1\}$$

Free parameter $\lambda \in [0, 1] \rightarrow$
continuous family.

Lean: `familyII_OK` $\forall (n, m, s, \lambda)$
passing screens.

λ -Families A–E (verified)

5 families at $n=5$, each covers ∞
values.

$$\Sigma_Z(\lambda) = \sum_i \langle Z_i \rangle^2 \in [\tfrac{1}{3}, 1]$$

What universality buys

- One 100-line file $\rightarrow$ **all** instantiations
- Proves **structure**, not cases
- Classification complete for $|C_0| \leq 4$

Open: analytical families for **distance 3?**

Needs universal proofs over $\mathbb{Q}(\sqrt{\cdot}, i)$ with
amplitude parameters — significantly
harder.

Outlook: Beyond BD_{16}

BD_{16} at $n=7$ is the **first step**. The framework extends naturally:

Other BD groups

$\text{BD}_{12}\text{--}\text{BD}_{36}$ ($m=6\text{--}18$)

Same pipeline; number field adapts to ω_m .

Zhang–Zeng: dozens of $((7, 2, 3))$ candidates

$n=8$: $T^{1/4}$

BD_{64} ($m=32$): transversal $\pi/16$.

Intersection test still effective.

Exceptional groups

$2I$ ($|G|=120$): 5th level

$2T$ ($|G|=24$)

$2O$ ($|G|=48$)

Two-layer architecture still applies.

Long-term vision: a formal library of quantum codes with **machine-verified** fault-tolerance properties — AI discovers, Lean certifies.

Summary

Summary

Distance 2

- Full SSLP pipeline in Lean 4
- Residue-shift screen $\Rightarrow d \geq 2$ automatic
- 14,116 codes certified
- Analytical families: one proof $\rightarrow \infty$ codes

Codebase

$\sim 4,300$ lines Lean 4, 0 sorry
github.com/LionSR/lean-qec

Distance 3 (new)

- Intersection test filters most of 210 KL
- GPT-5.2 $\rightarrow$ closed-form amplitudes
- Exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, i)$
- All 10 feasible BD_{16} vectors verified; 2 Lean no-go proofs

What's next

Batch 14K codes distance-2 automation $\text{BD} \rightarrow \text{BD}$ $n=8$ 21/27/20

Thank you

arXiv:2510.20728 · github.com/LionSR/lean-qec